→ paradigm shift in statistical thinking

"Probability approach in econometrics" (1944)

classical interpretation ↔ frequentist interpretation ← Probabilistic approach (includes Bayesian statistics) — Trygve Haavelmo

prob = principle of indifference (coins/die etc) ∴ natural symmetry of n problem (toy problem for classical view)

$$P(x) = \lim_{n_t \to \infty} \frac{n_x}{n_t}$$

n_x ← number of trials; n_t ← total number of trials

↳ representation of world in terms of "degrees of belief" (Bayesian prior)

UP504

Review Part 1

Bayes' Theorem

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

Random variables

1) Discrete random variables and expectations (observational data)

categorical/qualitative: nominal, ordinal, dichotomous

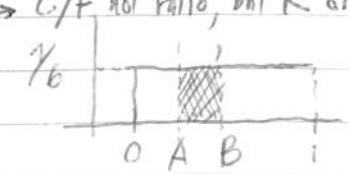
- Discrete random variable
- Continuous random variable

→ uniform distribution

quantitative variables: interval, ratio (but with 0 meaning none of that variable)

interval: differentiation preserves units

ratio → C/F not ratio, but K or height

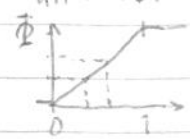


2) Expectation of discrete random variable

Expectations operator

$$E(X) = x_1 p_1 + x_2 p_2 + \dots + x_n p_n = \sum_{i=1}^{n} x_i p_i$$

with CDF



↳ $E(X)$ is the population mean, recall that $\sum p_i = 1$

Example: St. Petersburg's Paradox (Bernoulli) → fluid dynamics: speed-pressure relationship

Proposal to pay $\$2^{n-1}$ if T at n^{th} toss

$$\hookrightarrow E(X) = \tfrac{1}{2}\cdot\$1 + \tfrac{1}{2}\cdot\tfrac{1}{2}\,\$2 + \tfrac{1}{2}\cdot\tfrac{1}{2}\cdot\tfrac{1}{2}\,\$4 \dots$$

$$= \tfrac{1}{2}\,\$1 + \tfrac{1}{4}\,\$2 + \tfrac{1}{8}\,\$4 + \dots$$

$$= \sum_{i=1}^{\infty} \$_i \tfrac{1}{2} = \infty$$

Example $E(X) = \frac{1}{6} \cdot 1 + \frac{1}{6} 2 + \ldots + \frac{1}{6} 6 = 3.5$

with $\mu_{DIE} = 3.5$ and pair of dice $\mu_X = 7$

Generally, we can write $E(X) = E[g(X)]$

$= g(x_1)p_1 + \ldots + g(x_n)p_n$

Expectation operator

Expectations rules

① $E[X+Y+Z] = E[X] + E[Y] + E[Z]$

② $E[bX] = bE[X]$

③ $E[b] = b$

Example: $E[a+bX] = a + bE(X)$

Variance of discrete variable

$$Var(X) = \sigma_X^2 = E[(X-\mu_X)^2]$$
$$= (x_1-\mu_X)^2 p_1 + (x_2-\mu_X)^2 p_2 + \ldots$$
$$= \sum_{i=1}^{n} (x_i-\mu_X)^2 p_i$$

Exercise: $\sigma_X^2 = E[(X-\mu_X)^2] =$

$$= E[(X-\mu_X)(X-\mu_X)] =$$

$$= E[X^2 - 2\mu_X X + \mu_X^2] =$$

$$= E[X^2] + E[-2\mu_X X] + E[\mu_X^2]$$

$$= E[X^2] - 2\mu_X E[X] + \mu_X^2$$

$$= E[X^2] - 2\mu_X^2 + \mu_X^2 = E[X^2] - \mu_X^2$$

Fixed and random components of variables

$X = \mu_X + u$ eg. measurement error, common trends

with $E(u) = E[X - \mu_X] = E[X] - \mu_X = 0$

$\therefore$ because all variation in X is due to u, $\sigma_X^2 = \sigma_u^2$ since

$$\sigma_X^2 = E[(X-\mu_X)^2] = E[X^2] - \mu_X^2 = E[u^2]$$

$$\sigma_u^2 = E[(u-\mu_u)^2] = E[u^2]$$

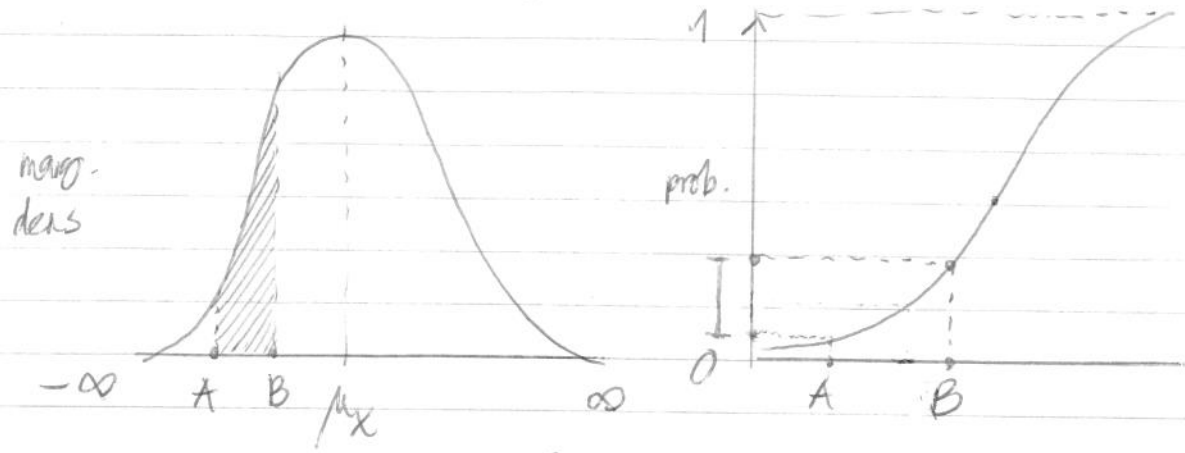
Research design:

4 key questions when doing "good quantitative research" (Angrist & Pischke, 2010)

1. what is the causal relationship of interest (→ importance of the "counterfactual")
2. what ~~would~~ ideal experiment would capture causal effect?
3. what is the identification strategy (how is observational data used) (3)
4. what is the mode of inference (microdata, clustered groups etc.)

Continuous Random variables

PDF with $f(x) \equiv F'(x)$ CDF



$$\underline{Pr}[A \le X \le B] = \int_A^B f(x)\,dx = F(B) - F(A) \quad , \quad \therefore \text{CDF} \equiv \int_{-\infty}^{x} f(x)\,dx = \int_{-\infty}^{x} F'(x)\,dx$$

Expected value of continuous random variable

$$E[X] = \int X f(x)\,dx = \mu_X$$

Similarly:

$$E[(X-\mu_X)^2] = \sigma_X^2 = \int (X-\mu_X)^2 f(X)\,dx$$

* Moments of a distribution : Expectations of random variable raised to power;
3. skewness, 4. kurtosis

4 $$m_k(X) = \int x^k f(X)\,dx \rightarrow \frac{\text{central moments}}{\mu_k \equiv E[X - E(X)]^k} = \int (X-\mu_X)^k f(x)\,dx$$

Covariance and correlations

$$cov(X,Y) = \sigma_{XY} = E[(X-\mu_X)(Y-\mu_Y)]$$

Independence between X and Y iff

$$E[g(X)h(Y)] = E[g(X)]E[h(Y)] \therefore \sigma_{XY} = 0$$

because recall that $E[X-\mu_X] = E[X] - \mu_X = 0$

Covariance rules

1. $Y = W + V$; $cov(X,Y) = cov(X,W) + cov(X,V)$
2. $Y = bZ$; $cov(X,Y) = b\,cov(X,Z)$
3. $Y = b$; $cov(X,Y) = 0$

Variance rules

1. $Y = W + V$; $var(Y) = var(W) + var(V) + 2cov(V,W)$
2. $Y = bZ$; $var(Y) = b^2 var(Z)$
3. $Y = b$; $var(Y) = 0$
4. $Y = V + b$; $var(Y) = var(V)$

Correlation (unlike covariance, it does not rely on measurements of X and Y)

$$\rho_{XY} = \frac{\sigma_{XY}}{\sqrt{\sigma_X^2 \sigma_Y^2}}$$ ∴ expression is unit free

Samples and Estimators ($X_1 \rightarrow x_1$) dual structure of variable

Generally population characteristics are unknown ∴ engage in sampling to estimate mean, variance by collecting realisations of variables

Estimator ≡ rule for estimating an unknown population parameter, defined in terms of the sample observations $(X_1, X_2 \ldots X_k)$

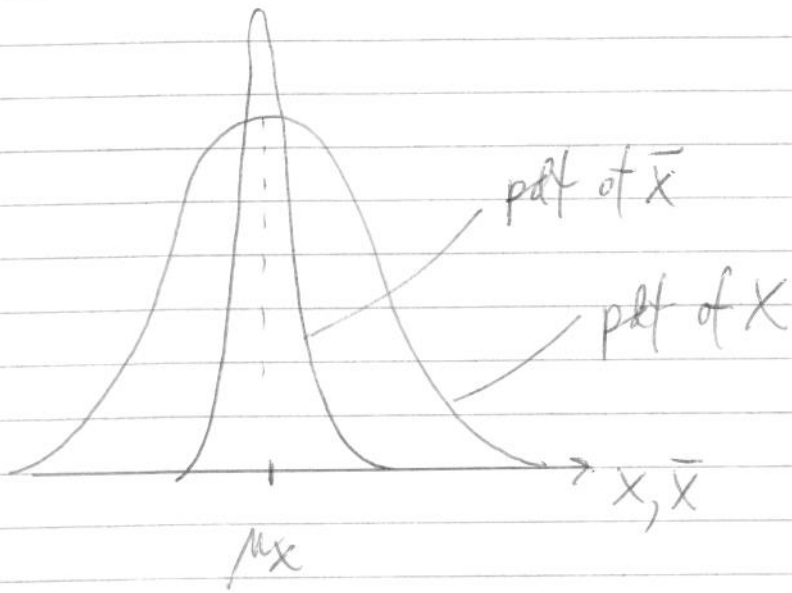
↳ estimate is generated on basis of actual sample $(x_1, x_2 \ldots, x_n)$

∴ Repeated sampling yields different estimates

↳ estimators are a combination of random variables which makes them a special case of random variables

Eg. Sample mean ($\bar{X}$) as estimator of population mean (μ_X)

$$\bar{X} = \tfrac{1}{n}(X_1 + X_2 + \dots + X_n) = \frac{1}{n}\sum_{i=1}^{n} X_i \equiv \mu_X$$



↳ $\sigma^2_{\bar{X}} = \dfrac{\sigma^2_X}{n}$ due to dual structure of variables with

(i) potential distribution of obs in sample

(ii) sample mean before and after realisation of sample

$$\sigma^2_{\bar{X}} = var\left(\frac{1}{n}(X_1 + X_2 + \dots + X_n)\right) =$$

$$= \frac{1}{n^2}\left(var(X_1) + var(X_2) + \dots var(X_n)\right)$$

$$= \frac{1}{n^2}\left(\sigma^2_X + \sigma^2_X + \dots + \sigma^2_X\right) = \sigma^2_X/n$$

→ OLS estimator provides estimate of conditional mean of Y given the values of X; ie

$$E[Y|X] = \beta_0 + \beta_1 X$$

<u>Estimator properties</u> (Finite sample vs. asymptotic properties)

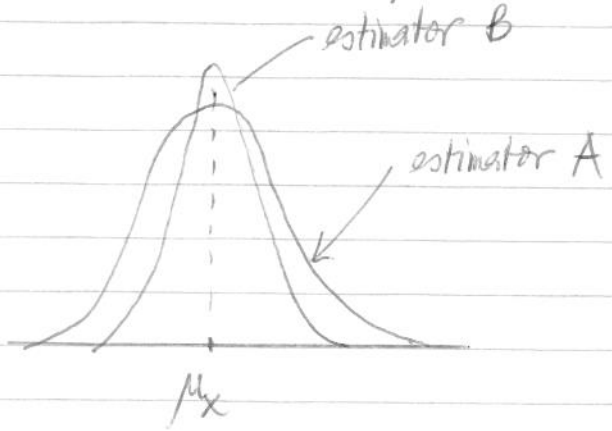
<u>UNBIASEDNESS</u>, estimate not systematically different from true value

$$E(\bar{X}) = E\left[\frac{1}{n}(X_1 + X_2 + \ldots + X_n)\right]$$

$$= \frac{1}{n}E[X_1 + X_2 + \ldots X_n] = \frac{1}{n}[\mu_x + \mu_x + \quad + \mu_x]$$

$$= \mu_x$$

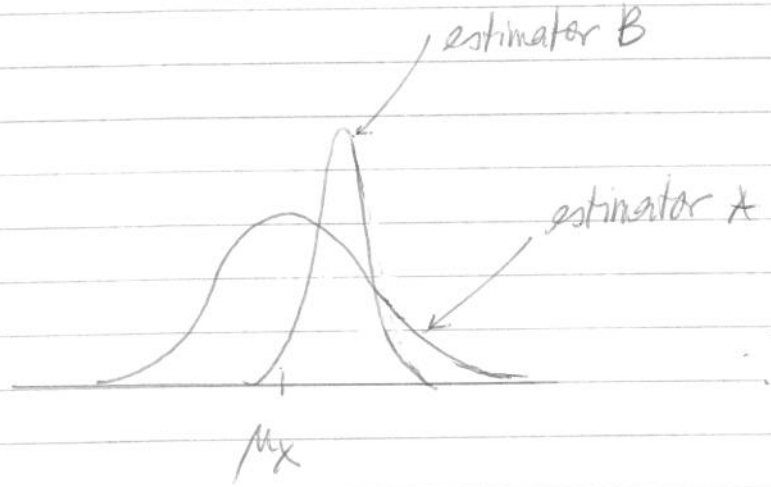
<u>EFFICIENCY</u>; estimate produced is "close as possible to true value" (ie. minimal dispersion around mean)



→ can show that sample variance of <u>mean</u> is the smallest of its kind.

∴ of unbiased estimators for mean, sample average is also most efficient.

↳ efficiency and bias are relative concepts that induce a <u>possible trade-off</u>



How do we choose either A or B? Pick a loss function such as MSE

$$\text{MSE of estimator} = E\left[(\hat{\theta}-\theta)^2\right]^*$$

$$= \underbrace{\text{variance of estimator}}_{①} + \underbrace{\text{bias}^2}_{②}$$

* <u>CONSISTENCY</u> $\hat{\theta}$ is a consistent estimator of θ if $\hat{\theta} \xrightarrow{p} \theta$ as $n \to \infty$

$$* \; E\left[(\hat{\theta}-\mu_{\hat{\theta}}+\mu_{\hat{\theta}}-\theta)^2\right] = E\left[(\hat{\theta}-\mu_{\hat{\theta}})^2 - 2(\hat{\theta}-\mu_{\hat{\theta}})(\mu_{\hat{\theta}}-\theta) + (\mu_{\hat{\theta}}-\theta)^2\right]$$

$$= \underbrace{E\left[(\hat{\theta}-\mu_{\hat{\theta}})^2\right]}_{①} - 2(\mu_{\hat{\theta}}-\theta)\underbrace{\cancel{E\left[\hat{\theta}-\mu_{\hat{\theta}}\right]}}_{=0} + \underbrace{E\left[(\mu_{\hat{\theta}}-\theta)^2\right]}_{②}$$

Estimators of sample variance, covariance, correl.

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2 \rightarrow$$ unbiasedness proof ITE, p. 79

$$s_{xy} = \frac{1}{n-1} \sum^{n} (X_i - \bar{X})(Y - \bar{Y})$$

Normal distribution:

$$\phi = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left\{\frac{X-\mu}{\sigma}\right\}^2}$$

$X \sim N(\mu, \sigma^2)$ and standardised $N(0,1)$

Sample size and estimator properties: ASYMPTOTICS

	Reject H_0	Do not reject
H_0 true	$\boxed{\text{SIZE}}$ TYPE I	$1-\alpha$
H_0 false	$1-\beta$ $\boxed{\text{POWER}}$	β TYPE II

UP 504 <u>Review Part 2</u>

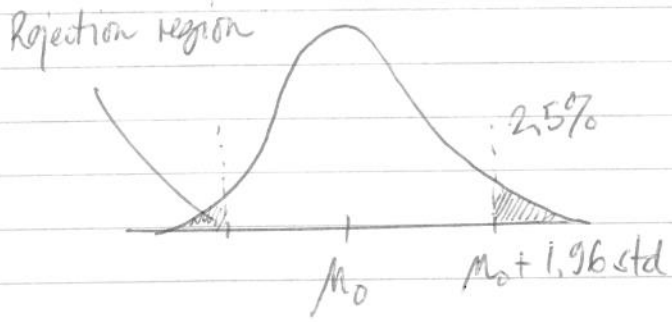
Hypothesis Testing

$H_0: \mu = \mu_0$ vs. $H_1: \mu \neq \mu_0$

If null hypothesis is true, $\bar{X}$ will be distributed $N \sim (\mu_0, \sigma^2/n)$

Type I error: Reject when H_0 true

Type II error: Non-rejection when H_0 false



<u>Note</u>: 6σ covers full distribution

$$z = \frac{\bar{X} - \mu_0}{s.d.(X)} \gtrless \pm 1.96 \quad \text{(actually observed value } \hat{z}\text{)}$$

$\sigma^2_{\bar{X}} \to S^2_{\bar{X}}$

s.d is not actually known; use estimate s.e. $\boxed{S_{\bar{X}} = \left[\frac{1}{n-1}\sum(X_i - \bar{X})^2\right]^{1/2}}$

<u>P-values</u>: Go beyond simple reject / not reject duality since significance levels are <u>essentially arbitrary</u>

↳ marginal significance level: "Greatest level for which a test based on $\hat{z}$ fails to reject the null."

→ p-values give exact value of type I error

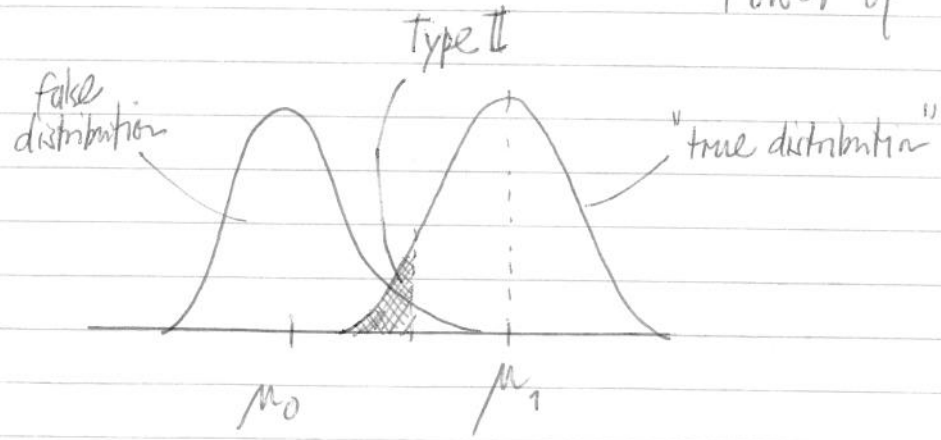
$$p(\hat{z}) = 2(1 - \Phi(|\hat{z}|))$$

$\hat{z}_1 = 2.02$
$\hat{z}_2 = 5.77$ } reject both at 5%, but magnitude not directly visible.

↳ $P_1 = 4.34\%$; $P_2 = 0.00000079\%$

∴ Rejection region ≡ significance level / size of test

Probability of rejecting false H_0 = 1 − prob (Type II)
= Power of test



↳ The closer μ_0 and μ_1, the smaller the power of a test

Confidence interval [around a POINT ESTIMATE]
vs
INTERVAL ESTIMATE

$$\bar{X} - 1.96\,\sigma_{\bar{X}} \leq \mu \leq \bar{X} + 1.96\,\sigma_{\bar{X}} \qquad (95\%\ CI)$$

↳ But since $\sigma_{\bar{X}}$ is not observed, we must use estimate and corresponding critical value from the t-distribution

$$t = \frac{\bar{X} - \mu_0}{s.e.(\bar{X})}$$

IMPORTANT: RATIO OF 2 RANDOM VARIABLES FOLLOWS t-DISTRIBUTION

One-sided versus two-sided testing usually depends on context of investigation:

(i) One-sided test common for policy evaluation; eg. treatment vs. effect

(ii) $H_0: \mu = 0$ economic theory that determines sign of parameters.

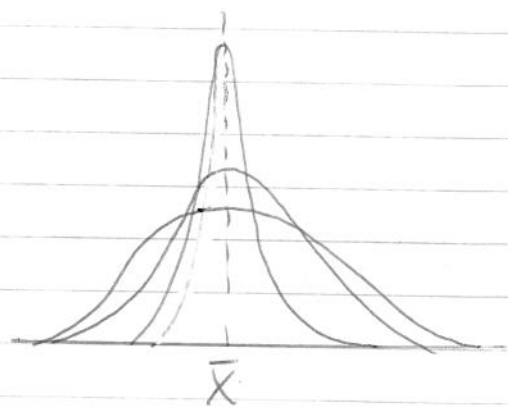
Probability limits

$$\lim_{n \to \infty} P(|Z - \alpha| > \varepsilon) \to 0$$

$$\therefore \text{ plim } Z = \alpha$$

$$\lim_{n \to \infty} P(|\bar{X} - \mu_x| > \varepsilon) \to 0$$

$$\therefore \text{ plim } \bar{X} = \mu_x$$



Conditions for consistency

1. Estimator of population characteristic possesses plim
   $\therefore$ distribution collapses into spike for large n
2. Spike is located at true value

↳ Unbiasedness as sufficient condition; not necessary since even biased estimators can be consistent at large sample sizes.

Finite vs. large sample properties (asymptotic properties)

↳ in an analytical context plim rules are much more powerful than E[ ] which often require independence and other strong assumptions.

Plim rules

(i) $\text{plim}(X+Y+Z) = \text{plim}(X) + \text{plim}(Y) + \text{plim}(Z)$

(ii) $\text{plim}\, bX = b\, \text{plim}\, X$

(iii) $\text{plim}\, b = b$

(iv) for $Z = XY$, $\text{plim}\, Z = \text{plim}\, X \cdot \text{plim}\, Y$

(v) for $Z = X/Y$, $\text{plim}\, Z = \text{plim}\, X \cdot (\text{plim}\, Y)^{-1}$

(vi) $\text{plim}\, f(X) = f(\text{plim}\, X)$

Central Limit Theorem

$\bar{X}$ will be normally distributed, even if X is not normally distributed, given a <u>large enough</u> sample size.

↳ depends on underlying distribution:

(i) for uniform, $n > 10$ sufficient

(ii) for lognormal, $n > 100$